Dr Deepak Parashar

Professor of Biostatistics | Clinical Trials Methodologist | AI for Health
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Profile

Biostatistician and data scientist with over 15 years of research and teaching experience in medical statistics, clinical trial methodology, and AI for health. Currently Deputy Lead of the Turing–Roche Strategic Partnership at The Alan Turing Institute and Visiting Professor of Biostatistics, Bioinformatics & Biomathematics at Georgetown University (Lombardi Comprehensive Cancer Center, Washington DC). Author of 50 peer-reviewed publications spanning Nature Machine Intelligence, Nature Communications, JCO Precision Oncology, Statistics in Medicine, Pharmaceutical Statistics, and Annals of Oncology, with research concentrated on biomarker-driven adaptive enrichment designs, master protocols, type I error control, personalised uncertainty quantification, and geometric methods in data science. Experienced educator with over 20 years of university-level teaching — from first-year modules to doctoral supervision — with three completed PhDs (MRC iCASE partnerships with Roche and AstraZeneca), one current PhD student at Georgetown University, and sustained curriculum leadership at Warwick Medical School.

Current Positions

Deputy Lead, Turing–Roche Partnership 2024 – 2026

The Alan Turing Institute for Data Science & Artificial Intelligence, London

- Part of senior leadership team delivering cutting-edge research on Disease and Treatment Heterogeneity in Precision Medicine.
- Manage and mentor postdocs working on: Structured Missingness, Fairness in Integrated Risk Tools, Conformal Prediction.
- Collaborate with UK and US data scientists; liaison with Roche/Genentech (San Francisco).

Visiting Professor of Biostatistics, Bioinformatics & Biomathematics 2025 –

Georgetown University, Washington DC, USA

- Strengthening methodology collaboration with biostatisticians, clinical oncologists, and computational biologists at the Lombardi Comprehensive Cancer Center and Innovation Center for Biomedical Informatics.
- PhD thesis co-advisor: Confirmatory Clinical Trial Designs for Evolutionary Guided Precision Medicine for Cancer.

Visiting Researcher 2025 –

MRC Biostatistics Unit, University of Cambridge

Research Interests

Personalised Medicine and Adaptive Designs in Cancer Clinical Trials · Drug Development · Advanced Statistical and AI Methods for Health Data Analytics · Geometric Methods in Data Science

Education & Qualifications

Years	Institution	Qualification
2010	*University of Warwick*	PG Certificate in Academic & Professional Practice (Maths)
1996–2000	*Robert Gordon University, Aberdeen*	PhD Mathematics (conferred March 2001)
1994–1996	*International School for Advanced*	Pre-doctoral course in Mathematical

	Studies (SISSA), Trieste	Physics (First Class)
1993–1994	*University of Cambridge*	Certificate of Advanced Study in Mathematics (Honours) 1994; Master of Advanced Study, conferred 2011
1991–1993	*University of Delhi*	MSc Physics (First Class)
1988–1991	*University of Delhi*	BSc (Honours) Physics (First Class)

Teaching & Supervision

Over 20 years of university teaching in mathematics and medical statistics at all levels. Fellow of the Higher Education Academy. Published scholarly article on mathematics education. Module design, exam setting, PhD and MSc supervision, and curriculum leadership.

Modules Taught (select)

- Swansea (2004–2005): Vector Calculus (2nd year maths).
- Warwick Mathematics Institute (2005–2008): Differential Equations (1st year, 300+ students); Quantum Groups (4th year M.Math).
- Cambridge (2010–2014): Survival Analysis, Sample Size & Modern Trial Design for MPhil in Translational Medicine; Statistics for CRUK Cambridge PhD researchers.
- Warwick Medical School (2014–2025): MB ChB statistics lectures & clinics; MSc Dentistry; Masters in Public Health; MRC Warwick DTP; BSc Health & Medical Sciences.
- UHCW Integrated Academic Training: Hypothesis Testing & Regression, Power & Sample Size, Survival Analysis for trainee doctors (annual).
- STEM Connect MSc in Digital Health, Data & Diagnostics (2023–2025): co-lead for new core module 'AI and Machine Learning for Diagnostics'.
- MB ChB SocPop Co-Leadership (2020–23): Co-theme Lead for 'Social and Population Perspectives' across all 4 years of the MB ChB programme.

PhD Supervision — Completed

- Omar Abdullah: 'Venous thromboembolism in bladder cancer: Scope of the problem and patients' experiences.' CARA Fellowship, 2017–2021. Awarded 2021.
- Ben Lanza: 'Statistical Design and Analysis of Clinical Studies using Personalised Healthcare under Biomarker Uncertainty.' MRC iCASE studentship (Roche), 2019–2023. Awarded 2024.
- Amelia Thompson: 'Quantitative Decision-Making in Clinical Drug Development Incorporating Biomarkers.' MRC iCASE studentship (AstraZeneca), 2019–2023. Awarded 2024.

PhD Supervision — Current

- Patrick Roney: 'Confirmatory Clinical Trial Designs for Evolutionary Guided Precision Medicine for Cancer.' Georgetown University, 2025–. (Co-advisor)

Selected Master's Project Supervision

- MathSys CDT: 'Biomarker-stratified design of cancer clinical trials' — highest marks in cohort year (2015).
- MPH: mammographic image quality — led to publication with major international impact, criteria adopted in European breast cancer screening guidelines (2016).
- MRC DTP: 'Support Vector Machines in stratifying patient populations based on cancer biomarkers' — awarded distinction, led to publication (2019).
- MRC DTP: 'Retrospective evaluation of adaptive designs fitted to precision oncology clinical trial data' (2021).

Previous Employment

Academic — Medical Statistics / Data Science (2009–2025)

Associate Professor in Systems Biology & Oncology 2019 – 2025
Warwick Medical School & Warwick Cancer Research Centre, University of Warwick

Assistant Professor in Systems Biology & Oncology 2014 – 2019
Warwick Medical School & Warwick Cancer Research Centre, University of Warwick

Turing Fellow (Visiting) 2018 – 2023
The Alan Turing Institute for Data Science & Artificial Intelligence

Senior Statistician 2013 – 2014
Cancer Research UK Cambridge Institute, University of Cambridge

Biostatistician 2009 – 2013
Cambridge Cancer Trials Centre & MRC Biostatistics Unit Hub in Trials Methodology Research, University of Cambridge

Academic — Pure Mathematics (2000–2009)

Associate Fellow / Lecturer 2008 – 2009
University of Warwick

Visiting Professor 2007 – 2008
IHES, Paris and Hausdorff Research Institute for Mathematics, Bonn

Visiting Research Fellow 2006 – 2007
Max Planck Institute für Mathematik, Bonn

Assistant Professor 2005 – 2007
Warwick Mathematics Institute, University of Warwick

Postdoctoral Research Fellow / Lecturer & Tutor 2002 – 2005
University of Wales Swansea

Visiting Researcher 2003 – 2004
Università di Roma 'Tor Vergata'

Postdoctoral Research Fellow 2000 – 2002
Max Planck Institute for Mathematics in the Sciences, Leipzig

Membership of Learned Societies

Fellow, The Higher Education Academy (Awarded 2016)

Honorary Director, Natural Computing Applications Forum (NCAF), 2011–2019

Early career: Graduate Member, Institute of Physics · Junior Member, Isaac Newton Institute for Mathematical Sciences · Fellow, Cambridge Commonwealth Society · Member, New York Academy of Sciences · Member, Edinburgh Mathematical Society

Publications

50 papers in peer-reviewed journals; co-edited one book. Approximate % contribution shown; highlighted entries are particularly significant. Listed by discipline, with medical statistics and AI publications leading.

Biostatistics, Medical Statistics and Artificial Intelligence

1. Accelerated BEP: a phase I trial of dose-dense BEP for intermediate and poor prognosis metastatic germ cell tumour. Y. Rimmer et al. British Journal of Cancer 105 (2011) 766–772. [15%]
2. Ultrasound elastography as an adjuvant to conventional ultrasound in the preoperative assessment of axillary lymph nodes in suspected breast cancer. K. Taylor et al. Clinical Radiology 66 (2011) 1064–1071. [15%]

3. A randomised study evaluating the use of pyridoxine to avoid capecitabine dose modifications. P.G. Corrie et al. British Journal of Cancer 107 (2012) 585–587. [20%]
4. Timing of Surgery Following Neoadjuvant Long Course Chemoradiotherapy in the Management of Locally Advanced Rectal Cancer. I. Panagiotopoulou et al. Clinical Oncology 24 (2012) e195. [40%]
5. **The diagnostic value of white cell count, C-reactive protein and bilirubin in acute appendicitis and its complications. I. Panagiotopoulou et al. Annals of the Royal College of Surgeons 95 (2013) 215–221. [40%]**
6. The Cambridge post-mastectomy radiotherapy (C-PMRT) index: a practical tool for patient selection. M. Mukesh et al. Radiotherapy and Oncology 110 (2014) 461–466. [40%]
7. Diagnostic accuracy of pre-operative computed tomography used alone to detect lymph node involvement at radical nephrectomy. S.S. Connolly et al. Scandinavian Journal of Urology 49 (2015) 142–148. [40%]
8. Surgical margin length and location affect recurrence rates after robotic prostatectomy. H. Dev et al. Urologic Oncology 33 (2015) 109.e7–109.e13. [40%]
9. **The UK experience of a treatment strategy for paediatric metastatic medulloblastoma (Milan Strategy). S. Vivekanandan et al. Pediatric Blood & Cancer 62 (2015) 2132–2139. [40%]**
10. Neo-adjuvant Long Course Chemoradiotherapy for Rectal Cancer: does time to surgery matter? I. Panagiotopoulou et al. International Surgery 100 (2015) 968–973. [40%]
11. Comparing the use and interpretation of PGMI scoring to assess the technical quality of screening mammograms in the UK and Norway. M. Boyce et al. Radiography 21 (2015) 342–347. [40%]
12. **Type I error control in biomarker-stratified clinical trials. D. Parashar et al. Trials 2015, 16(Suppl 2):O84. [70%]**
13. Investigating the ability of multiparametric MRI to exclude significant prostate cancer prior to transperineal biopsy. E.M. Serrao et al. Can Urol Assoc J 9 (2015) E853–858. [40%]
14. Transperineal prostate biopsies for diagnosis of prostate cancer are well tolerated: a prospective study using patient-reported outcome measures. K. Wadhwa et al. Asian Journal of Andrology 18 (2016), 1–5. [40%]
15. **An optimal stratified Simon two-stage design. D. Parashar et al. Pharmaceutical Statistics 15 (2016) 333–340. [70%]**
16. **Detection of colorectal dysplasia using fluorescently labelled lectins. J.C-H. Kuo et al. Nature Scientific Reports 2016; 6:24231. [30%]**
17. **Mortality Among Men with Advanced Prostate Cancer Excluded from the ProtecT Trial. T.J. Johnston et al. European Urology 2016, 71: 381–388. [40%]**
18. **Mammographic image quality in relation to positioning of the breast: A multicentre international evaluation. K. Taylor et al. Radiography 2017; 23: 343–349. [40%] — Editors' Choice Award; criteria incorporated in European guidelines for breast cancer screening.**
19. Integrated Care in Prostate Cancer (ICARE-P): Nonrandomized Controlled Feasibility Study. V. Nanton et al. J. Med. Internet Res. 2017;6(7):e147. [15%]
20. **On the need to adjust for multiplicity in confirmatory clinical trials with master protocols. N. Stallard et al. Annals of Oncology 2019. [15%]**
21. Promoting integrated care in prostate cancer through online holistic needs assessment. A.L. Clarke et al. Supportive Care in Cancer 2020; 28: 1817–1827. [15%]
22. **Do Support Vector Machines play a role in stratifying patient population based on cancer biomarkers? B. Lanza and D. Parashar. Arch Proteom Bioinform. 2021; 2: 20–38. [50%]**
23. Venous Thromboembolism Rate in Patients with Bladder Cancer According to the Type of Treatment: A systematic review. O.R. Abdullah et al. Cureus (Nature) 2022; 14(3): e22945. [15%]
24. **Efficiency of a randomized confirmatory basket trial design constrained to control the false positive rate by indication. L. He et al. Statistical Methods in Medical Research 2022; 31(7): 1207–1223. [15%]**
25. **Unlocking multidimensional cancer therapeutics using geometric data science. D. Parashar. Scientific Reports (Nature) 2023; 13: 8255. [100%]**
26. Effect on maximal mouth opening in children with spinal muscular atrophy treated with onasemnogene abeparvovec. N. Beri et al. Archives of Disease in Childhood (BMJ) 2023; 108:866–867. [20%]
27. Onasemnogene abeparvovec gene therapy for spinal muscular atrophy: A cohort study from UAE. V. Mundada et al. Muscle and Nerve 2024;70(4):808–815. [20%] (Statistical Lead)

28. Generalized Evolutionary Classifier for Evolutionary Guided Precision Medicine. M.D. McCoy et al. JCO Precision Oncology 2025: 9:e2300714. [30%] (Project co-lead)

29. Personalised Uncertainty Quantification in Artificial Intelligence. T. Chakraborti et al. Nature Machine Intelligence 2025; 7: 522–530. [10%] (Turing–Roche partnership leadership)

30. Generating crossmodal gene expression from cancer histopathology improves multimodal AI predictions. S. Dey et al. Nature Communications 17, Article 259 (2026). [10%] (Turing–Roche partnership leadership)

Statistics Education

31. Challenges in mathematics and statistics teaching underpinned by student-lecturer expectations. D. Parashar. Eur. Jour. Sci. Math. Education 2 (2014) 202–219. [100%]

32. Immortal Time Bias: Zeroing in on time-zero. D. Parashar and J. Wang. PSI SIG on Real-World Data (peer-reviewed), 2024. [50%]

Commissioned Reports

33. Epidemiological modelling and prediction. Co-Leads: S. Denaxas (UCL), D. Parashar (Warwick). The Alan Turing Institute, June 2021.

34. Pathogenesis and virus evolution, vaccines and clinical trials. Lead: J. Weatherall (AstraZeneca); co-authored by D. Parashar et al. The Alan Turing Institute, June 2021.

35. UK-India AI Opportunities Policy Paper: A Way Forward for the UK-India Joint AI Centre. Janjeva, T. Chakraborti, D. Parashar. The Alan Turing Institute, January 2026. [Invited by UK FCDO]

Manuscripts Under Review / In Preparation

- Towards Multimodal Artificial Intelligence Assurance. T. Chakraborti, C. Burr, D. Parashar et al. Communications in AI & Computing. [40%] (Turing–Roche partnership leadership)
- A Proof-of-Concept Clinical Trial Design for Evolutionary Guided Precision Medicine for Cancer. D. Parashar et al. medRxiv 2025; doi:10.1101/2025.05.23.25328210. Submitted to Scientific Reports. [60%] (Project co-lead)
- Dynamic Precision Medicine: Navigating the Evolutionary Landscape of Cancer to Reshape 21st Century Oncology. O. Sverdlov, D. Parashar, R.A. Beckman et al. [10%] Target: 2026.

Pure Mathematics (see also Appendix)

15 papers in peer-reviewed mathematics journals and proceedings, plus one co-edited Springer volume (Quantum Groups and Noncommutative Spaces, 246 pp., 2010). Full list in Appendix.

Preprint: R-matrices for the adjoint representations of Uq(so(n)). D. Parashar and B.W. Westbury. arXiv:0906.3419, 2009.

Reviewing Activity

Journals: Lett. Math. Phys., Crelle's Journal, Statistics in Medicine, Biometrics, BMJ Health & Care Informatics, Patterns (Cell Press), PLOS One, Scientific Reports, npj Digital Medicine, Communications Medicine, Data & Policy, among others.

Grants & Panels: US National Science Foundation, NIHR Health Technology Assessment Programme, MRC Hubs for Trials Methodology Research, European Research Agency Personalised Medicine Review Panel, Cancer Research UK, Yorkshire Cancer Research, MRC AI for Biomedical & Health Research Panel, MRC Clinical Academic Research Partnerships Panel, UKRI AI Innovation to Accelerate Health Research Panel.

Research Grants & Contracts

No.	Date	Project Title / Details / Duration	Funding Body	PI / CoI	Other CoI / PI	Total Awarded
1	2024	Deputy Lead, Turing–Roche Partnership 2024–26	The Alan Turing Institute for Data Science and AI	PI	N/A	0.4 FTE
2	2023	Data Science	Warwick's	PI	N/A	£5,000

		and AI for Health Duration: 1 year; 2023–24	STEM Grand Challenge seed funding			
3	2023	Prognostic marker study designs for precision health data Duration: 2 years; 2023–25	Warwick's International Partnership Fund	PI	CoI: Prof. Bhramar Mukherjee (U. Michigan / Yale)	£10,000
4	2023	Statistical methods for biomarker-driven clinical trial designs Duration: 1 year; 2022–23	Warwick's Health GRP	PI	N/A	£3,000
5	2021	Turing Fellowship Duration: 2 years; 2021–23	The Alan Turing Institute for Data Science and AI	PI	N/A	Expenses
6	2020	Detection of metastatic axillary sentinel lymph nodes using ultrafast, super-resolution dual-contrast enhanced ultrasound imaging in patients with breast cancer Duration: 4 years	NIHR i4i	CoI	PI: Prof. Mengxing Tang (Imperial College London)	£975,000
7	2019	Quantitative Decision-Making in Clinical Drug Development Incorporating Biomarkers MRC DTP IBR studentship conversion to iCASE Duration: 3 years	AstraZeneca	PI	N/A	£19,800
8	2019	Statistics driven cancer research: From maths to medicine Public engagement at 2019 British Science Festival Duration: 6 months	Wellcome–Warwick Quantitative Biomedicine (WQBP) Public Engagement Fund	PI	N/A	£3,000
9	2019	Novel Trial Designs for Dynamic Precision	The Royal Society International Exchanges	PI	CoI: Prof. Robert Beckman (Georgetown	£11,959

		Medicine in Cancer Bilateral visits for US collaboration Duration: 4 years	Award		University, USA)	
10	2019	Workshop on Statistical Trends in Personalised Healthcare	The Alan Turing Institute for Data Science and AI	PI	N/A	£3,300
11	2018	Turing Fellowship Duration: 2 years; renewed 2020–21	The Alan Turing Institute for Data Science and AI	PI	N/A	£13,195
12	2018	Project on Design Animation for Stratified Cancer Trials	Warwick Cancer Research Centre	PI	N/A	£2,000
13	2018	Statistical Design and Analysis of Clinical Studies using Personalised Healthcare under Biomarker Uncertainty MRC Industrial CASE PhD Studentship via Warwick IBR DTP Duration: 4 years	MRC and Roche Pharmaceuticals	PI – Project Academic Lead	Warwick IBR Doctoral Training Programme	£107,940+ (Roche: £21,600; MRC: £86,340)

Previous PI Grants as Early Career Researcher

Period	**Funding Body / Grant**	**Award**
2012	*Conference on Clinical Biostatistics — national workshop grant, Natural Computing Applications Forum*	£3,000
2007	*Workshop on Quantum Groups and Noncommutative Geometry, Max-Planck-Institut für Mathematik, Bonn*	€20,000
2006–2007	*Max-Planck-Institut für Mathematik (Bonn) — 12-month Visiting Fellowship*	€36,000
2003–2004	*European Union Fifth Framework — Quantum Spaces & Noncommutative Geometry (Rome/Swansea)*	€26,000
2002–2005	*Royal Commission for the Exhibition of 1851 Fellowship, Swansea University (One of 6 fellowships awarded worldwide across all science &*	£50,000

	engineering)	
2000–2004	*Further grants: European Science Foundation, Max-Planck Society, JSPS*	Various

NHS Trusts — Lead Statistician on Clinical Trials

Trial / Study	Description	Phase / Type
ABEP	*Accelerated BEP chemotherapy for intermediate and poor prognosis metastatic germ cell tumour*	Phase I
CAPP-IT	*Role of pyridoxine in controlling hand-foot syndrome in patients with advanced colorectal and breast cancer*	Phase III RCT
IMRT	*Cambridge Breast Intensity Modulated Trial — dose inhomogeneity correction and cosmetic outcome*	RCT
CAP01	*Pharmacokinetics of capecitabine in patients post proximal pancreatico-duodenectomy*	Phase I PK
Elastography	*Ultrasound elastography as adjunct to conventional ultrasound for axillary lymph node assessment in suspected breast cancer*	Pilot study
CRT	*Delay from neoadjuvant chemoradiotherapy to surgery and pathological complete response in rectal cancer*	Retrospective
Appendicitis	*Diagnostic accuracy of WCC, CRP and bilirubin for acute and perforated appendicitis*	Retrospective
PGMI	*Agreement in PGMI scoring for mammographic quality assessment across UK and Norway*	Observational
Design input	*Contributed to trial design for: ABEPII, BUBBLE, 18F-PET, RT-TMZ*	Various

Select Recent Invited & Contributed Talks

- Technology and Innovation in Healthcare, FCDO (invited to Heads of British Overseas Territories). London, November 2025.
- Biomarker-driven cancer trials: Regulatory and methodology advances. University of Michigan Cancer Data Science Center, Ann Arbor, USA, May 2025.
- A proof-of-concept clinical trial design for evolutionary guided precision medicine for cancer. ASA Joint Statistical Meeting, Portland OR, USA, August 2024.
- Visualising Master Protocols. PSI Conference, Amsterdam, June 2024.
- Emerging trends in regulatory considerations of AI/ML applications in clinical trials. 7th ISBS Symposium, Baltimore MD, USA, March 2024.
- Biomarker-based trial designs: Methodology and regulatory issues. Boehringer-Ingelheim Pharma, Biberach, Germany, June 2023.
- Precision Medicine Trial Designs: Is there hope? PSI Conference, London, June 2023.

- A predictive biomarker enrichment design for Phase II oncology trials. Adaptive Designs Workshop, Basel, April 2023.
- Enrichment designs with predictive biomarkers. PSI Biomarkers ESIG webinar (400+ attendees), January 2023.
- Designing clinical trials for dynamic precision medicine in cancer. Georgetown University, Washington DC, USA, October 2022.
- Leveraging real-world data for time-to-event endpoints. PSI Conference, Gothenburg, June 2022.

Impact, Outreach & Engagement

International Academic Impact

- Led collaboration with Georgetown University (funded by Royal Society International Exchanges Award) and University of Michigan Ann Arbor (Warwick International Partnership Fund).
- Appointed Co-mentor, Regeneron Talent Search of USA.
- Health Mentor, UK-India AIXcelerate programme; reviewing and mentoring teams for prototype Health AI solutions at the Global AI Impact Summit 2026.
- Co-organised Track 1.5 Dialogue: UK-India AI Opportunities, The Alan Turing Institute, November 2025.

Industry & NHS

- Research collaborations with Roche and AstraZeneca — each sponsored an MRC iCASE studentship, 2018–2023.
- Sustained partnerships with NHS trusts since 2009: Addenbrooke's, University Hospitals Birmingham, UHCW Coventry, Maidstone & Tunbridge Wells Hospitals Trust.

Public Engagement & Policy

- Invited to scientific receptions at Buckingham Palace — invitation of HRH Prince Philip (2006) and HRH Princess Anne (2016).
- Delivered sold-out lecture 'The maths of personalised medicine' at the 2019 British Science Festival (with 3D animation; attracted pre- and post-event press coverage).
- Co-author of UK-India AI Opportunities Policy Paper invited by UK Foreign & Commonwealth Development Office.
- Appointed Fellow, Warwick Institute of Engagement.
- Honorary Director, Natural Computing Applications Forum (NCAF), 2011–2019.

Collegiality, Leadership & Management

Strategic Leadership

- Deputy Lead, Turing–Roche Partnership: senior leadership position overseeing research themes on Structured Missingness, Fairness in Integrated Risk Tools, and Conformal Prediction; managing 3 postdocs.
- Lead, STEM Connect Research Group in Data Science and AI for Health — first collaborative interdisciplinary research group spanning Warwick Medical School, Statistics, WMG, Computer Science, and Centre for Interdisciplinary Methodologies.
- Co-theme Lead for Cancer Trials, Warwick Cancer Research Centre.
- Executive Group Member, Health Global Research Priority Programme, University of Warwick.
- Lead for 'Methodology' team, PSI Special Interest Group on Real-World Data.
- Co-lead, UK-India AI Policy Paper for the UK and Indian Governments on behalf of The Alan Turing Institute.

Institutional Contributions

- Contributed to the creation of the Warwick Cancer Research Centre.

- Led Warwick's first successful MRC iCASE institutional bid and renewal; Project Lead for 2 iCASE PhD projects.
- Statistical Lead on Warwick's NIHR i4i national grant.
- Led Warwick's bid for the Decovid programme at The Alan Turing Institute.
- Member, Warwick HR Excellence in Research Working Group (revised Concordat contributions).

Appendix: Early Career Pure Mathematical Research

My PhD (Aberdeen, 1996–2000) addressed multiparameter quantum groups and the problem of duality using both the R-matrix approach and the geometric approach. I generalised these frameworks and obtained several rigorous results on duality between coloured quantum groups and quantum algebras.

As a Postdoctoral Fellow at the Max Planck Institute Leipzig (2000–2002), I developed the theory of 'coloured quantum groups' parametrised by continuously varying parameters. As a Royal Commission for the Exhibition of 1851 Fellow at Swansea (one of 6 worldwide), I formulated differential calculus on the coloured quantum plane — 'coloured quantum spaces' with connections to integrable systems and knot theory.

At Warwick Mathematics Institute (2005–2007) and during visiting positions at Max Planck Bonn, Hausdorff Research Institute, and IHES Paris (2006–2008), I developed coloured Yang–Baxter operators as solutions to the spectral-parameter-dependent QYBE, with applications in Baxter's solvable models and quantum computing. I co-organised an international conference and co-edited a Springer book on Quantum Groups and Noncommutative Spaces.

Select Early Career Conference Talks

- Quantum Groups and Quantum Spaces, New Mexico State University, USA, January 1997.
- On the algebro-geometric structure of coloured quantum groups, The Fields Institute, Toronto, January 2001.
- Some algebro-geometric aspects of biparametric and coloured quantum groups, IPAM, UCLA, November 2001.
- Coloured Yang-Baxter operators, North Carolina State University, Raleigh, USA, May 2005.

Workshop Organisation

Workshop on Quantum Groups and Noncommutative Geometry, Max-Planck-Institut für Mathematik, Bonn, August 2007 (co-organised with Prof. Matilde Marcolli).

References available on request.